

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Description: Cerium(III) 2-ethylhexanoate, 49% in 2-ethylhexanoic acid, Ce 12%
Cat No. : 40451
Molecular Formula C24 H45 CeO6

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

Details of the supplier of the safety data sheet

Importer

Fisher Scientific Korea
D5,D6, Incheon Airport Logistics Complex
150, Gonghangdong-Ro 296 Beon-Gil
Jung-Gu, Incheon
Tel: +82-1661-9555
Fax: +82-2-2023-0603

Supplier

Thermo Fisher Scientific Chemicals, Inc.
30 Bond Street
Ward Hill, MA 01835-8099

E-mail address Chem.KR@thermofisher.com

Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Reproductive Toxicity

Category 1

Environmental hazards

Based on available data, the classification criteria are not met

Label Elements

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Signal Word

Danger

Hazard Statements

H360 - May damage fertility or the unborn child

Precautionary Statements

Prevention

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P280 - Wear protective gloves/protective clothing/eye protection/face protection

Response

P308 + P313 - IF exposed or concerned: Get medical advice/attention

Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

Other Hazards

Toxicity to Soil Dwelling Organisms

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

NFPA

Health
0

Flammability
0

Instability
0

Physical hazards
-

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	Common Name	CAS No	Index No	Weight %
2-Ethylhexanoic acid	2-Ethylcapronic acid	149-57-5	KE-13740	50 - 60
Cerium 2-ethylhexoate	No information available	56797-01-4	KE-13744	40 - 50

SECTION 4: FIRST AID MEASURES

Description of first aid measures

General Advice

If symptoms persist, call a physician.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

Ingestion

Clean mouth with water and drink afterwards plenty of water.

Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

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Self-Protection of the First Aider Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

Extinguishing media

Suitable Extinguishing Media

Carbon dioxide (CO₂). Powder. Water spray. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion.

Extinguishing media which must not be used for safety reasons

No information available.

Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂), Cerium oxide.

Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required.

Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system.

Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid

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ingestion and inhalation.

Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

Specific End Uses

Use in laboratories.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS No	Korea	ACGIH TLV	OSHA PEL
2-Ethylhexanoic acid	149-57-5	Not listed	TWA: 5 mg/m ³	Not listed
Cerium 2-ethylhexoate	56797-01-4	Not listed	Not listed	Not listed

Component	CAS No	European Union	The United Kingdom	Germany
2-Ethylhexanoic acid	149-57-5	Not listed	Not listed	Not listed
Cerium 2-ethylhexoate	56797-01-4	Not listed	Not listed	Not listed

ACGIH - Biological Exposure Indices

Component	CAS No	ACGIH - Biological Exposure Indices
2-Ethylhexanoic acid	149-57-5	Not listed
Cerium 2-ethylhexoate	56797-01-4	Not listed

Exposure Controls

Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Wear safety glasses with side shields (or goggles)

Hand Protection

Protective gloves

Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Personal protective equipment

Respiratory Protection

Use only those certified by the Korea Occupational Safety and Health Administration.

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

Recommended Filter type:

Multi-purpose/ABEK conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls

Prevent product from entering drains

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SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) Liquid Viscous liquid

Odor No information available

Odor Threshold No data available

pH Not applicable

Melting Point/Range No data available

Softening Point No data available

Boiling Point/Range No information available

Flash Point > 110 °C / > 230 °F **Method -** No information available

Evaporation Rate No data available

Flammability (solid,gas) Not applicable Liquid

Explosion Limits No data available

Vapor Pressure 23 hPa @ 20 °C

Vapor Density No data available (Air = 1.0)

Specific Gravity / Density 1.08 g/cm³ @ 20 °C

Bulk Density Not applicable Liquid

Water Solubility Immiscible

Solubility in other solvents No information available

Partition Coefficient (n-octanol/water)

Component	CAS No	log Pow
2-Ethylhexanoic acid	149-57-5	2.7
Cerium 2-ethylhexoate	56797-01-4	No data available

Autoignition Temperature No data available

Decomposition Temperature No data available

Viscosity No data available

Explosive Properties No information available

Oxidizing Properties No information available

Molecular Formula C₂₄ H₄₅ CeO₆

Molecular Weight 569.74

SECTION 10: STABILITY AND REACTIVITY

Reactivity None known, based on information available

Chemical Stability Store under inert atmosphere.

Possibility of Hazardous Reactions

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Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

Conditions to Avoid None known.

Incompatible Materials Oxidizing agent.

Hazardous Decomposition Products Carbon monoxide (CO). Carbon dioxide (CO₂). Cerium oxide.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

Information on expected route of exposure

Inhalation Not an expected route of exposure.
Ingestion May be harmful if swallowed.
Eyes Avoid contact with eyes.
Skin Avoid contact with skin.

Information on Health Hazards

(a) acute toxicity;

Oral Based on available data, the classification criteria are not met
Dermal Based on available data, the classification criteria are not met
Inhalation Based on available data, the classification criteria are not met

Toxicology data for the components

Component	CAS No	LD50 Oral	LD50 Dermal	LC50 Inhalation
2-Ethylhexanoic acid	149-57-5	LD50 = 1600 mg/kg (Rat)	LD50 = 1140 mg/kg (Rabbit)	No data available
Cerium 2-ethylhexoate	56797-01-4	500 (ATE)	1100 (ATE)	1.5 (ATE)

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available
Skin No data available

Component	CAS No	Test method	Test species	Study result
2-Ethylhexanoic acid	149-57-5	No data available	No data available	No data available
Cerium 2-ethylhexoate	56797-01-4	No data available	No data available	No data available

(e) germ cell mutagenicity; No data available

Component	CAS No	Test method	Test species	Study result
2-Ethylhexanoic acid	149-57-5	No data available	No data available	No data available
Cerium 2-ethylhexoate	56797-01-4	No data available	No data available	No data available

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(f) carcinogenicity; No data available

Component	CAS No	Test method	Test species / Duration	Study result
2-Ethylhexanoic acid	149-57-5	No data available	No data available	No data available
Cerium 2-ethylhexoate	56797-01-4	No data available	No data available	No data available

There are no known carcinogenic chemicals in this product

Component	CAS No	IARC	NTP	ACGIH	OSHA	UK
2-Ethylhexanoic acid	149-57-5	Not listed	Not listed	Not listed	Not listed	Not listed
Cerium 2-ethylhexoate	56797-01-4	Not listed	Not listed	Not listed	Not listed	Not listed

(g) reproductive toxicity; Category 1

Component	CAS No	Test method	Test species / Duration	Study result
2-Ethylhexanoic acid	149-57-5	No data available	No data available	No data available
Cerium 2-ethylhexoate	56797-01-4	No data available	No data available	No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

Other Adverse Effects

No information available.

Component	CAS No	EU - Endocrine Disruptors Candidate List	EU - Endocrine Disruptors - Evaluated Substances	Japan - Endocrine Disruptor Information
2-Ethylhexanoic acid	149-57-5	Not applicable	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable	Not applicable

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

Component	CAS No	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
2-Ethylhexanoic acid	149-57-5	LC50: = 70 mg/L, 96h (Pimephales promelas)	EC50: = 85.4 mg/L, 48h (Daphnia magna)	EC50: = 41 mg/L, 96h (Desmodesmus subspicatus) EC50: = 61 mg/L, 72h (Desmodesmus subspicatus)	EC50 = 110 mg/L 17 h EC50 = 670 mg/L 30 min
Cerium 2-ethylhexoate	56797-01-4	No data available	No data available	No data available	No data available

Persistence and degradability

Persistence

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary
Immiscible with water, May persist.

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Degradation in sewage treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

Bioaccumulative potential

May have some potential to bioaccumulate; Product has a high potential to bioconcentrate

Component	log Pow	Bioconcentration factor (BCF)
2-Ethylhexanoic acid	2.7	No data available

Mobility in soil

Spillage unlikely to penetrate soil The product is insoluble and sinks in water Is not likely mobile in the environment due its low water solubility.

Ozone Depletion Potential

Component	CAS No	Ozone Depletion Potential
2-Ethylhexanoic acid	149-57-5	Not listed
Cerium 2-ethylhexoate	56797-01-4	Not listed

Other adverse effects

No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport

Not Regulated

IATA

Not regulated

IMDG/IMO

Marine Pollutant

Not regulated
No hazards identified

Special Precautions for User

No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

Component	CAS No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	ISHL	AICS
2-Ethylhexanoic acid	149-57-5	KE-13740	X	205-743-6	X	X	-	X	X	X	X
Cerium 2-ethylhexoate	56797-01-4	KE-13744	X	260-386-3	X	X	-	X	X	X	X

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)

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2-Ethylhexanoic acid	149-57-5	Not applicable	Not applicable	Not applicable	Annex I - Y34
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
2-Ethylhexanoic acid	149-57-5	Listed	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable	Not applicable

Korean National Regulations

Component	CAS No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration
2-Ethylhexanoic acid	149-57-5	Annex 1 - KE-13740	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Annex 1 - KE-13744	Not applicable	Not applicable

Component	CAS No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
2-Ethylhexanoic acid	149-57-5	Not applicable	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable	Not applicable

Component	CAS No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
2-Ethylhexanoic acid	149-57-5	Not applicable	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable	Not applicable

Component	CAS No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance
2-Ethylhexanoic acid	149-57-5	Not applicable	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
2-Ethylhexanoic acid	149-57-5	Not applicable	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
2-Ethylhexanoic acid	149-57-5	Not applicable	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
2-Ethylhexanoic acid	149-57-5	Not applicable	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids

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				When Wet			
2-Ethylhexanoic acid	149-57-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

Control Parameters

Component	CAS No	Korea	ACGIH - Biological Exposure Indices
2-Ethylhexanoic acid	149-57-5	Not listed	Not listed
Cerium 2-ethylhexoate	56797-01-4	Not listed	Not listed

US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

Component	CAS No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
2-Ethylhexanoic acid	149-57-5	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable

CERCLA

Not applicable

Component	CAS No	CERCLA Extremely Hazardous Substances RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
2-Ethylhexanoic acid	149-57-5	Not applicable	Not applicable	Not applicable
Cerium 2-ethylhexoate	56797-01-4	Not applicable	Not applicable	Not applicable

GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Danger.

H360D - May damage the unborn child.

P201 - Obtain special instructions before use. P280 - Wear protective gloves/protective clothing/eye protection/face protection.

P308 + P313 - IF exposed or concerned: Get medical advice/attention.

SECTION 16: OTHER INFORMATION

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

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Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Prepared By

Health, Safety and Environmental Department

Creation Date

15-Apr-2015

Revision Date

17-Jun-2024

Revision Number

5

Revision Summary

New emergency telephone response service provider.

MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet